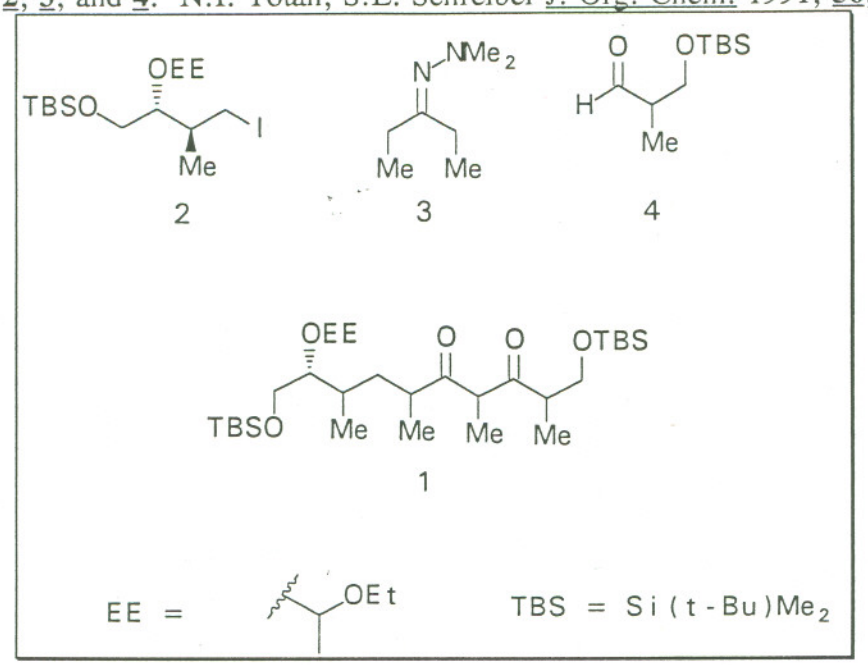
Dept. of Chemistry and Biochemistry

ORGANIC CUME

Dec 7th 1991

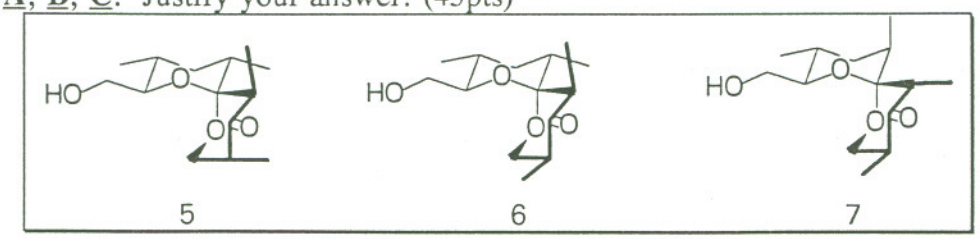
Schreiber has recently demonstrated a very elegant method to control the stereochemistry of easily epimerized (racemized) stereocenters. By a simple 4 step process 1,3-diketone **1** was prepared from **2**, **3**, and **4**. N.I. Totah, S.L. Schreiber J. Org. Chem. 1991, **56**, 6255.



1. Which stereocenters in compound **1** are readily epimerized under either acidic or basic conditions? (10pts)

2. Would you expect any one diastereomer of compound **1** to greatly predominate? Justify your answer. (15pts)

3. When **1** was deprotected, to give the trihydroxydiketone, and then treated with catalytic acid (0.5 mole equivalents of camphorsulphonic acid in MeOH) a mixture of three spiroketals were formed which could be equilibrated to give 3 spiroketals in the ratio of **A** : **B** : **C** 79 : 11 : 1. Assuming that the two six membered rings in these species are both "chair-like" and that the only conformations present in solution are represented by structures **5**, **6**, **7**, assign structures **5**, **6**, **7** to **A**, **B**, **C**. Justify your answer. (45pts)



4. These species could all be chromatographically separated. Resubjecting **B** to the catalytic acid conditions gave the same mixture (**A** : **B** : **C** 79 : 11 : 1). By way of mechanisms illustrate the sequence of events leading to the following isomerizations;

B ⟶ A

B ⟶ C

(30pts)